Natural Language Processing – 001

Data Collection

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Data Capturing

Origin

- Stack Overflow
- Pandas Documentation
- Matplotlib Documentation

Author(s)

Various users of the forum "Stack Overflow" and contributors (licensed under creative commons via Stack Overflow policy) to the documentation of Opensource API libraries (licensed under a permissive license).

Collection Method

Scraping using Python script and parsing with Beautiful Soup.

Raw Format

Raw HTML webpage (full pages are stored for now in case changes need to be made to our formatting step).

Our Formatting

After the raw data is collected, it will be converted into a consistent format so it can be used to train the model. Each example in the dataset will be made up of two components: a natural language prompt, and the Python code which performs the task. This data will be saved to a file for us to use later.

Read-In Method

Once the data is formatted into a file, we will start training the model by loading in the data from the file. The natural language prompt will be the model input, and the associated Python code will be the target output.

Normalization

We will need to do some normalization of the data between it being scraped and it being used as training data. First, we will extract the relevant text from the source, which will entail removing HTML formatting and isolating the prompt and code. Then, we will trim any unnecessary text like comments or notes that are not crucial to the data pair in question (keeping internal code comments to help with contextual intent). This should leave us with a clean pair of text and code that can be saved as mentioned above.

Annotation

We don't currently have any plans to add annotations to the data after normalizing it.

Trained Model

After training, the model will function as a natural language to Python code generator designed specifically for data analysis tasks. The model will take a user's question written in plain English as input and produce Python code that performs the requested operation on a dataset.

Model Information

The trained model will contain learned relationships between natural language questions and the Python code used to solve them. Through training, the model will learn how different phrases in user questions correspond to common data analysis operations performed using Python libraries such as Pandas or Matplotlib.

Decision-Making Features

The model will use the wording of the user's question to determine what type of analysis is being requested. Keywords such as "average," "sum," or "plot" help indicate the type of operation needed. The model will also use information from the uploaded dataset, such as column names, to generate Python code that works with the provided data.